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IT management and change at an energy firm: A teaching case on strategic alignment and business process management

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Abstract

This teaching case illustrates how important concepts and tools of information system management practice and literature are used in a young firm, challenged by strong growth and the need to integrate business processes and information technology systems. The case is typical of many smaller organizations that are substantially different from large firms. Among others, the firm does not have a dedicated information technology unit. As most of our information technology management knowledge stems from research in large companies, the teaching case demonstrates and combines many proven information technology management methods, and how they can be applied in smaller firms as well. Students will learn about strategic alignment, business process management, work system theory, business process management, and notation, and user resistance during information technology-induced change projects in small- and medium-sized organizations.

Keywords

Change, user resistance, business–information technology alignment, work system theory, business process management and notation, user acceptance

Introduction

TheCompany is a young, medium-sized firm in the energy sector with over 200 employees in 12 countries. It has grown rapidly in the past few years, and its processes have become larger and more complex. It is challenged by this impressive growth, global business, tight relations to larger parent companies, and changes in management. Consequently, from July 2018 to March 2019, a consultant was asked to analyze, update, and improve some of its business processes. This teaching case introduces useful tools, such as the business process management (BPM) cycle, work system theory (WST), business process management and notation (BPMN), and business IT alignment, and applies them to the case of TheCompany. By taking on the role of the consultant, students can familiarize themselves with these IT management tools, improve their analytical, presentation, information gathering, and group work skills (Ambrosini et al., 2010), and discuss the challenges of IT management in small- and medium-sized organizations that do not operate a dedicated IT unit. Moreover, the use of theoretical concepts in practice leads to a deeper understanding through active learning (Elliot, 2005).

The teaching case starts with the theoretical background that explains the core theoretical concepts (Patton and Appelbaum, 2003) and serves as a problem-solving IT-management "tool box." The main part consists of a summary of the company, interview, and observation results as well as process descriptions. The case focuses on a core process of the financial division and is based on a real-world project studied by the authors of this teaching case. Company information is underpinned with interviews

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and information from secondary data (Patton and Appelbaum, 2003). This information is the basis for the class discussion that is based on the tasks described in the final section. Teaching notes provide a structure for leading class discussions.

Theoretical background

This section introduces several IT management concepts related to the discussion of the case described in the following.

Firm-level

Strategic alignment. In 2018, business–IT alignment was one of the top two key issues among IT executives (Kappelman et al., 2019). Indeed, business–IT alignment, understood as the strategic, functional, and operational synchronization of a firm's IT and business units, has been among the most pressing IT management concerns for decades (Chan and Reich, 2007; Queiroz, 2017). The reason for this is that research and practical experience have consistently shown that IT, which is well synchronized with the business processes, is a main driver of organizational efficacy. Hence, managers try to enhance the maturity of alignment between IT and business (Luftman and Derksen, 2012).

Research indicates that business–IT alignment is particularly challenging for small- and medium-sized enterprises (SMEs), and, even more so, when the firm grows and expands its business activities. In those situations, organizations show less evidence of alignment while starting to introduce more formal planning processes to ensure an integrated vision for IT (Chan and Reich, 2007: 307). An additional SME characteristic that makes it difficult to directly transfer alignment research insights from larger firms is the lack of a separate IT unit in SMEs. This often leads to a decentralized IT infrastructure with business departments in charge of operating IT (Hussin et al. 2002; King et al., 2000) or a dependency on external consultants (Thong et al., 1994).

Business–IT alignment is about achieving coherence and synergies between business and IT (Hsiao and Ormerod, 2002). An increase in a firm's performance should be achieved by aligning these two domains on different levels (McLaren et al., 2011; Tallon, 2011), such as strategy and infrastructure (Henderson and Venkatraman, 1993; Nieminen and Pekkola, 2015). Various terms are used to describe alignment: fit (Gerow et al., 2014; Hussin et al., 2002; King et al., 2000), correlation (Palmer and Markus, 2000), fusion (Evans and Hoole, 2005), harmony (Chan and Reich, 2007; Luftman, 2000), integration (Henderson and Venkatraman, 1993), and linkage (Reich and Benbasat, 1996).

Strategic alignment has been mostly analyzed on a firm-level linking business strategy and IT strategy (Chan and Reich, 2007; Coltman et al., 2015; McLaren et al., 2011; Tallon et al., 2016). In line with strategy literature, business processes form the building blocks of a strategy (Queiroz, 2017; Ray et al., 2004; Stalk et al., 1992) and "are actions that firms engage in to accomplish some business purpose or objective" (Ray et al., 2004: 24). Based on the established differences between an intended strategy, referring to what a company wants to do, and a realized strategy, reflecting what a company actually does (Mintzberg and Waters, 1985), a shift from a firm-level to a process-level is observable in the recent literature of strategic business–IT alignment (Coltman et al., 2015; Queiroz, 2017; Tallon et al., 2016). Indicating a lack of IT support for business (Gerow et al., 2014; McLaren et al., 2011), misalignment on a process-level causes misalignment on a firm-level (Tallon, 2011). In addition, the implementation of strategic alignment into daily practice is reflected by operational alignment (Wagner et al., 2014; Wagner and Weitzel, 2012). This approach focusses on improving the relationships between and within business and IT departments (Schlosser et al., 2015; Wagner et al., 2006).

Strategic alignment model. Focusing on the firm-level, the renowned strategic alignment model (SAM) by Henderson and Venkatraman (1993) is described in the following. The SAM divides a firm into four domains: external versus internal and business versus IT (Figure 1). A *business or IT strategy* encompasses a scope, competencies, and governance regarding the respective competitive environment, mirrored in strategic partnerships, alliances, and make-or-buy decisions. While specifying the characteristics of a strategy, the point of view (business or IT) is crucial. *Organizational infrastructure and processes* reflect the firm's structure in terms of functional, divisional, or matrix organization, as well as the design of processes, and the skills of employees. The block *IT infrastructure and processes* is about the information system (IS) architecture covering technical infrastructure, such as applications as well as soft- and hardware, the IS process design referring to infrastructure in terms of maintenance or development, and IS skills of employees to effectively manage the IS infrastructure. In general, there are four potential cross-domain alignment paths a company can follow (Henderson and Venkatraman, 1993) (Figure 1). The first two approaches use the business strategy as a starting point:

1. *Strategy execution* (arrow a): The business strategy has been articulated (so-called anchor domain) and is the driver of both organizational (pivot domain) and IS design (impacted domain). This classic perspective is also known as business systems planning and enterprise modeling.
2. *Technology transformation* (arrow b): An increase in IT competencies should be achieved. The business strategy is also used as a basis, but first creates a fit

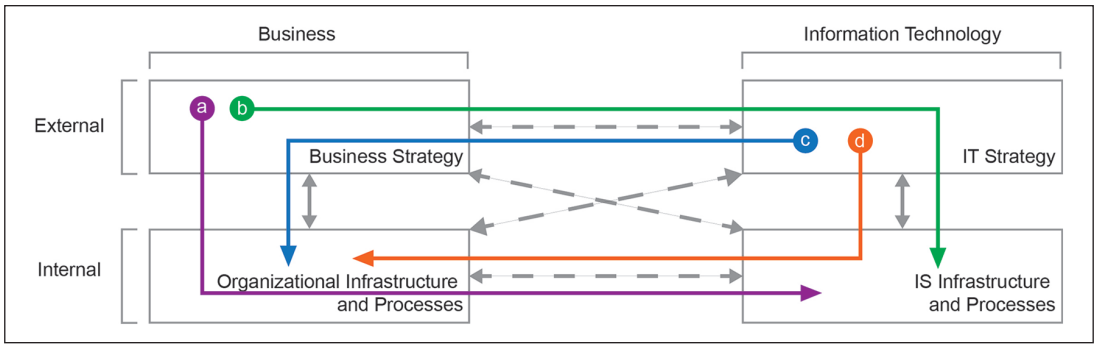


Figure 1. SAM with its cross-domain alignment paths. Source: reproduced with permission from Elsevier (Venkatraman et al., 1993).

Table 1. Reading list.

Concept	Authors	Year
Strategic alignment model	Henderson and Venkatraman	1993
Business process management	Dumas et al.	2013
Work system theory	Alter	2013
Business process management and notation	Object Management Group	2013
User resistance	Bovey and Hede	2001

with the IT strategy and adapts the internal IT afterward. This perspective is, therefore, not constrained by current organizational design but instead seeks to implement the best possible IT competencies.

The other two approaches use IT strategy as anchor domain:

1. *Competitive potential* (arrow c): To enable new business strategies using emerging technologies, the business strategy and the organizational structure are adapted to the IT strategy.
2. *Service level* (arrow d): For creating a world-class IS organization, an IT strategy has been formulated (anchor) and is implemented; the role of business strategy is indirect and viewed as providing the direction to stimulate customer demand.

Process-level

For process-level analysis and management, a variety of approaches are useful to cover the multifaceted nature of processes (Melao and Pidd, 2000): First, BPM and the BPM life cycle provided by Dumas et al. (2013) as an overarching framework; second, WST for an overall understanding of a company's work systems (WSs); third, BPMN to provide a notation for process visualization; and fourth, user resistance management to address resistance behavior that might occur during changes. Table 1 provides a reading list for these concepts; students should be familiar with them before analyzing and discussing the teaching case.

BPM. Dumas et al. (2013) describe the life cycle of BPM, aimed at continuously improving a firm's processes, in six phases (Figure 2). First, relevant business processes of a company are identified and set into relation, resulting in a visualized process architecture. Second, the current states of relevant processes are discovered and represented in as-is process models. Third, the models are analyzed, weaknesses are discovered and, if possible, quantified regarding their impact, and prioritized in terms of impact and manageability. Fourth, the chosen process is redesigned, and a to-be process model is created covering the objectives and issues determined before. Fifth, the to-be process model is implemented and set into action (change and implementation management). Sixth, the operational process is continuously monitored and evaluated in relation to the defined performance measures and objectives. Deviating behaviors, bottlenecks, and errors are identified and revised. Finally, the BPM life cycle starts again with the same or a new process.

WST. Following Alter (2013), WST provides an overview of WSs within a company. A WS is defined as "a system in which human participants and/or machines perform work (processes and activities) using information, technology, and other resources to produce specific products/services for specific internal and/or external customers" (Alter, 2013: 75), framed by strategy, environment, and infrastructure (cf. SAM). The idea is to create as-is and to-be snapshots. By comparing the snapshots, weaknesses can be discovered and potential interventions and actions to reach the to-be state can be derived.

WST analysis serves as a common basis for analysis and discussions among different stakeholders. A snapshot of a WS is a one-pager, including the six elements of a WS: the process and activities performed within, participants involved, information required, technologies used, services and products produced, and customers to be satisfied. If one element changes, effects might appear in the other elements and that

need to be considered in discussions (Alter, 2013). Figure 3 represents the structure of a snapshot.

BPMN. While WST describes processes with words, BPMN is a graphical notation with the purpose of process visualization and process implementation (Object Management Group, 2013). BPMN is widely used in theory and practice, and BPMN version 2.0.1 was elevated to international standard ISO/IEC 19510:2013 (International Organization for Standardization, 2013).

With its graphical elements, it is designed to coordinate the sequence of processes and messages that flow between participants in different activities. Serving as a common basis, it brings management, IT, and business employees together and provides richer information on who (the person, entity, or unit) does what and in what sequence. Aiming at a common understanding and communication, while being able to handle complexity inherent to business processes, a syntax and a semantic are defined. The graphical elements are arranged in a small set of categories: swim lanes, flow objects, data objects, connecting objects, and artifacts. A selection of elements necessary to fulfill the tasks of the given case is represented in Figure 4. Swim lanes build the foundation and are split into pools and lanes. Therein, flow objects describe the behavior of a process using events, activities, and gateways. Although it is conceivable that a step could span the border between adjacent swim lanes, each process step exists, by convention, within one and only one swim lane. Data objects are represented by a data object (e.g. a document, a data input, and a data output) or a data storage. Connecting objects link flow objects and/or other elements are described above. While sequence flows connect flow objects among themselves, associations connect data with flow objects. Artifacts are represented, for example, by text annotations. Providing these elements, a readable, flexible, and expandable representation of processes is made possible (Object Management Group, 2013; White, 2004).

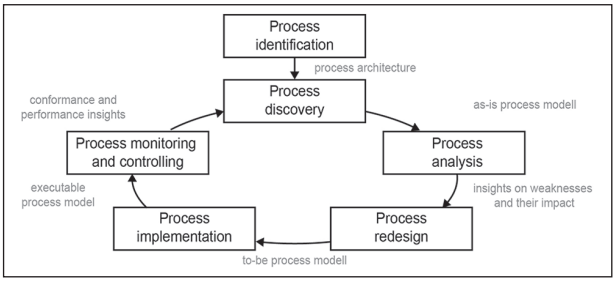


Figure 2. BPM life cycle. Source: reproduced with permission from Springer Nature (Dumas et al., 2013).

User resistance. During implementation, change management techniques contain interventions to adapt to the new

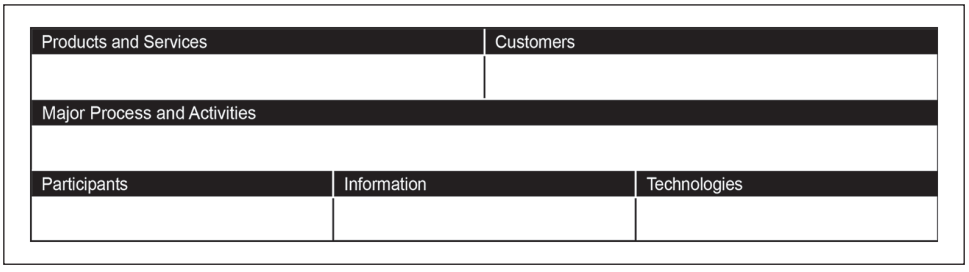


Figure 3. WS snapshot. Source: reproduced with permission from Elsevier (Alter, 2013).

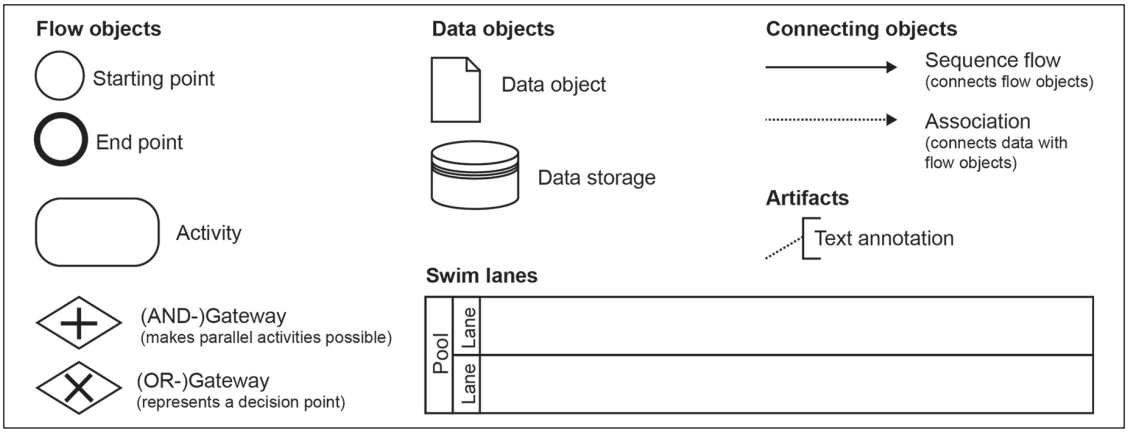


Figure 4. Graphical elements of BPMN.

way of working. However, change can induce different types of IS-related stress among employees (Laumer et al., 2016c; Maier et al., 2019). Thus, employee resistance to new systems or workflows is an often observed phenomenon that stands between the potential benefits of a new business process and their actual realization (Laumer et al., 2017). Research has identified possible causes of user resistance, including perceived threats by a new IS (Lapointe and Rivard, 2005), changes in work routines, or an associated feeling of loss of power (Laumer et al., 2016b; Lee and Joshi, 2017). Individuals differ in their willingness or ability to adapt to change (Laumer et al., 2016a). User resistance is mirrored in the behavior of individuals (Bovey and Hede, 2001; Mattke et al., 2018) and classified into four groups. The first differentiation is between active (originating action) or passive (non-action) resistance (Bovey and Hede, 2001). The second differentiation is between overt (open and expressive) or covert (concealed or hidden) resistance (Bovey and Hede, 2001). For each of the four arising categories, Bovey and Hede (2001) provide examples in the following. First, active and overt resistance is reflected in opposition, arguing, or obstructing. Second, active and covert user resistance shows as stalling, dismantling, or undermining. Third, passive and overt user resistance manifests as observing, refraining, and waiting. Fourth, passive and covert user resistance is depicted as ignoring, withdrawing, and avoiding. User resistance may cause delays, budget overruns, or underutilization of IS, and therefore should be avoided (Kim and Kankanhalli, 2009).

Company profile: TheCompany

Key facts

TheCompany is a business-to-business (B2B) player in the energy industry. Headquartered in Munich, Germany, TheCompany went live on April 2014 as a joint venture of two major players of different industries. The firm is dedicated to the global delivery of operational technology solutions linked with integrated IT to improve energy providers' performance. This is made possible by combining the products and operations technology knowledge of one of the parent companies with the capabilities in management and IT consulting, systems integration, and managed services of the other. By combining the arrays of services, TheCompany enables electric, gas, and water utilities to obtain greater reliability and efficiency of grid networks. Thereby, energy providers can execute advanced analyses and make fact-based decisions, due to the integrated view of their systems and data:

> We combine the best of two worlds of IT and OT [operational technology], integrating it into a common home with a focus on the energy utility market. Currently expanding in other industries but with a core on integration and further added values for our customers in terms of analytics data security and IoT [Internet of things]. (William Jones,[1] CFO)

The top-level management reflected the aspiration of the cooperation: the Chief Financial Officer (CFO) and Chief Executive Officer (CEO) formerly worked for one of the parent companies and the Chief Operating Officer (COO) for the other. Today, TheCompany operates globally and employs more than 200 individuals (Figure 5).

Industry and competitive environment

The evolution toward smarter grids leads to a big change in the energy sector. Challenging regulations increase, alternative and distributed energy sources emerge, and end-customer expectations evolve, while the network infrastructure ages. Only a few service providers are able to address these project-driven requirements. One of these players is TheCompany. Despite being a niche supplier, TheCompany has a strong background with its parent company, benefiting from a global footprint, products, and a secure financial background.

Vision and mission

TheCompany operates in a critical infrastructure environment with a responsibility toward customers and society. Therefore, TheCompany's vision for a new energy economy is smarter, stronger, greener, and more diverse. The mission is to work on a project basis with utilities to bridge the two worlds of operations and IT within the energy industry.

Company development

Since its founding in December 2014, TheCompany has been growing rapidly. The growth led to several organizational changes and developments, as shown in Figure 6. On the one hand, there were personnel changes in top-level management: a new COO, new CFO, and new CEO were hired in 2018. On the other hand, there were two major strategic changes: first, the acquisition of TheCompany by one of the parent companies in June 2018 and second, the move from one internal division of the parent company to another in April 2019. Due to the immense changes and rapid development, processes have grown over the years.

Organization and employees

TheCompany is led by a triangle of the CEO, William Jones as CFO, and a COO. With the power of around 200 experts spread over 12 countries, TheCompany has served more than 70 customers in projects across 15 countries by January 2019 (Figure 5). The international teams are based in 11 offices around the world, including Cologne, Amsterdam, Vienna, more than five offices in the United States, and a competency center in the Czech Republic. Acting globally and being medium-sized, at the same time, requires organizational skills due to global processes,

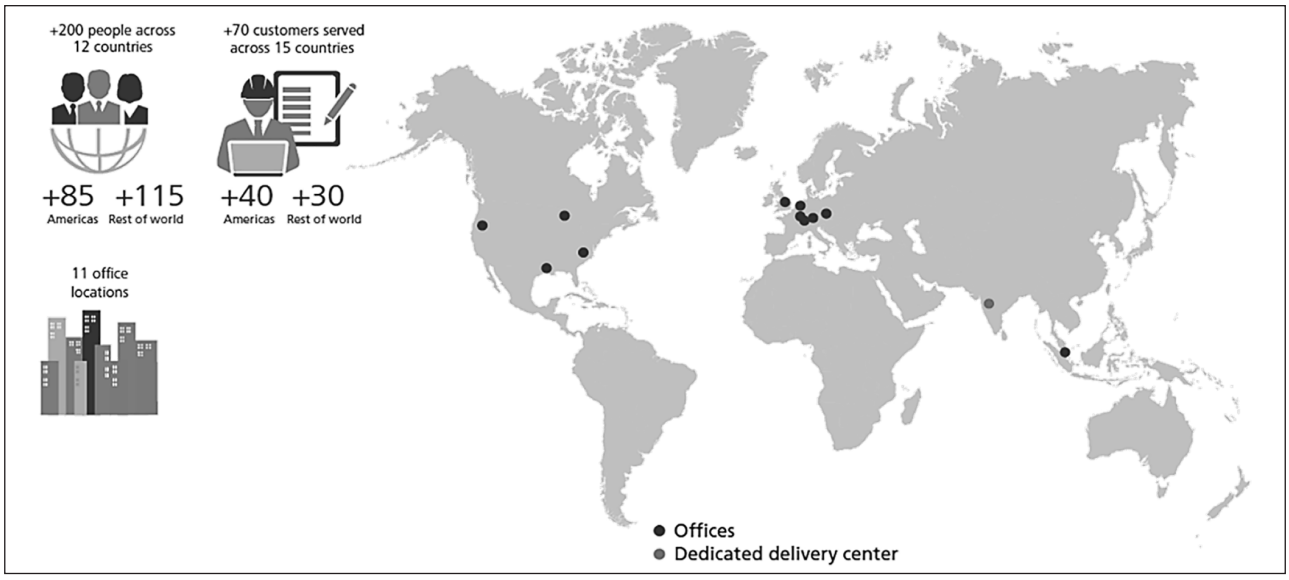


Figure 5. Global orientation of TheCompany.

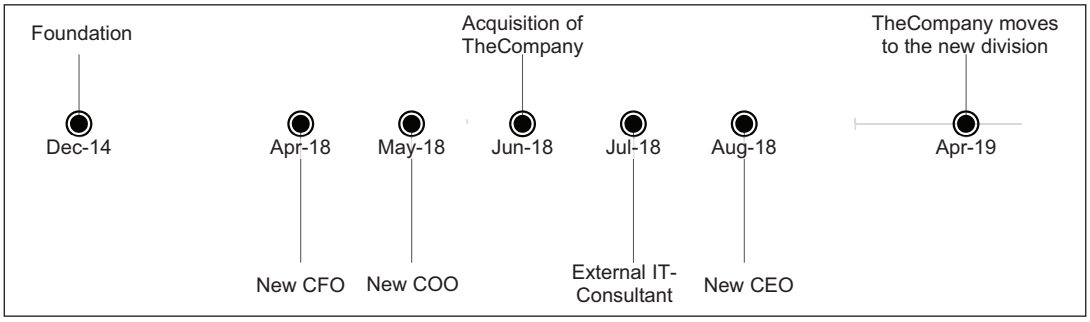


Figure 6. Organizational development at TheCompany.

strong online collaboration, and flexibility to cope with time differences. This makes the organizational structure and the way of working unique. To control the global business, TheCompany is divided into four legal entities: Germany (GER), Netherlands (NL), United States (US), and Austria (AT).

Finance department

The CFO is part of the top-level management. He leads the finance department, which employs about 15 employees. Corresponding to the organizational structure, the employees of the finance department are spread over four countries around the globe. In addition, a team in the Czech Republic supports the team in GER. In each country, there are teams of accountants and project controllers responsible for measurement, processing, and communication of the financial information referring to the legal entities. Considering all the combined information of all countries, the German team holds the global responsibility. To improve processes and business–IT alignment within the finance department, an external IT consultant was hired in July 2018. The tasks and interventions of the external IT consultant are described in the following.

Business analysis at TheCompany

From July 2018 to the end of March 2019, the external consultant gained insights into the finance department. He analyzed and optimized the different processes. In the following, interviews and analyses are presented that were made from July to October 2018. Then, the second phase is described that contains information about the implementation from November 2018 to the end of February 2019.

The external consultant and employees Max Müller and Lena Weber, accountants in GER, formed the change team that was initiated and sponsored by the CFO. Overall, the consultant invited seven employees. The CFO as part of the top-level management gave insights into strategic topics as well as his view on the processes. The other employees

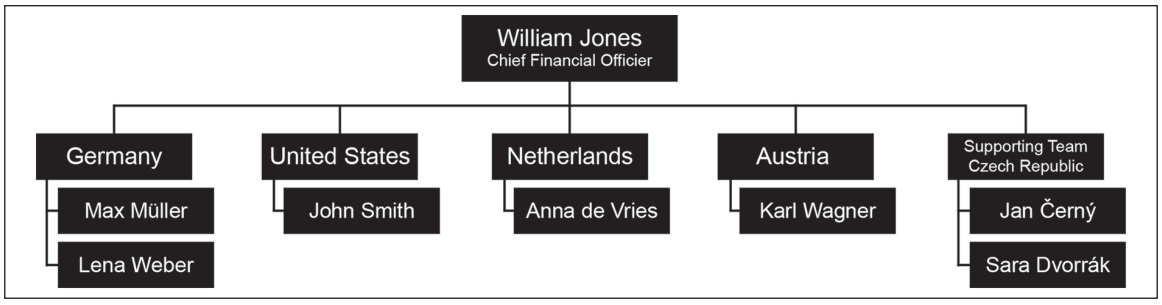


Figure 7. Organigram of employees interviewed.

supported the consultant in gathering information referring to the processes within the finance department. The employees were from AT, the NL, the US and GER, as well as from the supporting team in the Czech Republic (Figure 7). Due to the global dispersion of the employees involved, the main communication channels were Skype for Business, an internal chat and video telephony software, and e-mail.

John Smith, Anna de Vries, Karl Wagner, and Max Müller built the finance department of TheCompany from scratch. Lena Weber joined in October 2018. Before the project, the consultant was not in touch with TheCompany. In addition to the interviews, secondary data were gathered to gain an overall process understanding, encompassing reports, memoranda and meeting minutes, promotional material, press releases, internal documents, and interactions between employees as well as between the CFO and employees.

The results of the interviews and the secondary data are processed and summarized below. The first part encompasses the firm-level regarding the strategic orientation, the competitive environment, and developments as well as changes within the finance department. The second part looks at the process-level and particularly presents one process of the finance department. The focus is on the process to determine and report the sales, general, and administrative (SG&A) expenses. This process is applied on a monthly base to report the SG&A costs. By analyzing both, students should be enabled to apply the given theoretical concepts to a real-life scenario.

Firm-level analysis at TheCompany

The CFO described the competitive environment and the company's strengths as follows:

> The environment is very entrepreneurial, very fast, highly innovative and customer-focused. Moreover, we have a unique combination of operation and innovations technology on board with the ability to use the extremely strong product base and enriching it by the strong competencies that we have on it. We work in two different ways: We have our own business for interface design. We are responsible for the data incoming into the system, the project design, architecture development, data integration and app development. Depending on the project we work with different partners. (William Jones, CFO)

As displayed in the company profile, TheCompany ran through a major change due to the separation from one of the parent companies. In the course of this, an internal change was carried out. The CFO described the phase as follows:

> Today we are more independent, whereby a broader variety of value-adding partnerships is possible. I believe, a carefully designed and managed network of partners helps us to compete more effectively in our complex and dynamic markets. Nevertheless, both founding companies are our most important partners. However, we are open to working with other system integrators. In the future, we will profit from that. We are currently developing from an analytics and service provider towards a broader field of IoT. (William Jones, CFO)

TheCompany's development and changes in top-level management led to a new management style and changes within the organizational staffing of the finance team in GER. On the one hand, Max Müller who started the financial unit from scratch changed to one of TheCompany's parent companies in December 2018. On the other hand, one accountant and one project controller changed from one parent company to TheCompany in 2018. One of them is Lena Weber, taking over the tasks of Max Müller. In addition, another accountant was externally hired. All employees of the German financial unit were based in Cologne. Moreover, the CFO and the CEO used Cologne as a base:

> We implemented a finance department which is like a spider in the web. All information comes to that department. Talking about the finance team, we have the right people on board. Talking about other departments we still have vacancies. (William Jones, CFO)

To lead through this volatile phase, the new top-level management had strategy meetings on a regular basis and documented the new way forward in a strategy paper. Therein, the next development steps and the long-term development were outlined. Moreover, the roadmap incorporated

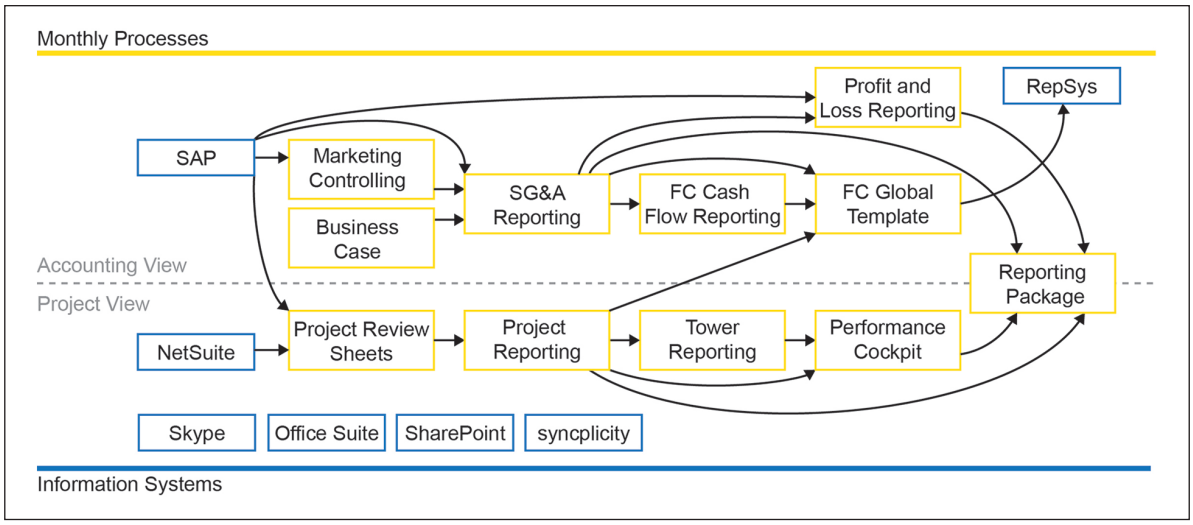


Figure 8. Process architecture of the monthly processes of the finance department.

defined key performance indicators for monitoring and controlling purposes. These served as an early warning system for the management. Besides the business scope, IT concerns were included in the strategy:

> IT infrastructure is essential in our business strategy. In particular, highly professional processes and IT systems are crucial. Covering the whole value chain from lead governance to financial processes. Of course, we need the right systems for the right processes. I think about sales management and control, forecasting, planning, reporting, SAP systems and booking, contract management. In general, you can say it is part of our business strategy, but it is on hold due to the divisional change. We want to look for synergies with the new division. But in the future, we have to change our systems. (William Jones, CFO)

As TheCompany did not have a CIO or respective IT unit, the responsibilities were split:

> Depending on which IT systems, there are different persons in charge. For instance, for the financial ones it is me, the CFO. (William Jones, CFO)

Being highly interwoven, TheCompany used soft- and hardware as well as the service desk of one of the parent companies. Within the finance department, the external consultant identified various ISs to fulfill everyday work: Microsoft Office, including Excel, Word, Outlook, and PowerPoint, Skype for Business, Syncplicity, Microsoft SharePoint, RepSys, NetSuite, and one Systems Applications and Products (SAP) system for each country (four SAP systems in total). RepSys is a general forecast and reporting system, and NetSuite covers the fields of business operations and customer relations in project-based concerns:

> Depending on the processes, for instance, we use NetSuite, SAP, RepSys. We are close to one of our parent companies that is why we share our systems partly. Additionally, we use a technology partner of our parent company supporting TheCompany's employees in daily IT issues. (William Jones, CFO)

Process-level analysis at TheCompany

At TheCompany, the fiscal year starts on the first of October, which is in line with the fiscal year of the parent company. The first quarter ends at the end of December and the second one at the end of March. In addition, the first of a month is called "U" and, for example, "U + 3" is the third workday of a month.

As mentioned in the company's profile, the finance department measures, processes, and communicates financial information. The consultant identified the finance department as being routine-driven by monthly and quarterly closings, as well as end-of-year reports. In addition, a split in two major areas of responsibilities within the finance department was observable (Figure 8): On the one side, there was the project-driven business requiring project reports, including overviews regarding current projects, order backlog, and project risk evaluations. On the other side, there was the accounting business providing actual, such as historical information, and forecasts (FC). Actuals and FC were reported monthly, for example, in a profit and loss report, a report referring to SG&A and FC cash flow report. Regardless of the type of financial report, or how the process was performed, every standard report was created in a final version once a month. Depending on the data availability, the final versions were available at different points in time. The interrelations and interdependencies between the processes raised the complexity and decreased the

transparency. Being the output of a process, a report could be deemed as the product of a respective process.

The external consultant visualized the processes of the finance department in a process architecture, shown in Figure 8. There, the different processes are represented with their names, their interrelations, and the IS in use. In addition, the processes were split into two perspectives: the project view required data from SAP and NetSuite, whereas the accounting view used SAP as input and RepSys as output for FC uploads. For international sharing and teamwork purposes, the employees used Microsoft Office, SharePoint, Skype for Business, and Syncplicity. Based on the interviews and secondary data, the consultant sketched the interdependencies of the main processes and IS within the finance department as the following.

Due to the rapid growth of the company, the CFO assumed that processes were performed in different variants depending on the respective employee:

> As a small start-up vehicle, many things have developed over time and have not been implemented within or based on a comprehensive plan. As we grow, we cannot handle this additional growth and complexity anymore. The processes should be easier to handle and faster. We want to standardize, automatize, and harmonize them globally, which means to have less error-prone processes caused by manual creation. Currently, there are significant differences within the processes of the four entities. (William Jones, CFO)

As the goal was to standardize and harmonize main processes, ad hoc analyses were not sufficient for the consultant. The expectations from the CFO were high:

> I expect higher efficiency, higher quality, faster data provision, more transparent reporting, lower costs and increased productivity. The process change team should be very structured, interdisciplinary in terms of overseeing the whole scope the change affects. They need to be highly engaged, strong in teamwork and strong collaborators and have of course really domain knowledge on board. (William Jones, CFO)

Besides that, the CFO asked to be kept in the loop during the entire project:

> I definitely want to see the progress milestones in-between to be able to see how it is done and I want to be involved in the discussions during the development. (William Jones, CFO)

From the CFO's perspective, there were two main processes: first, the process to report the SG&A expenses which were the main cost drivers within the company; second, the project reporting encompassing data referring to current projects and project backlogs. One major request of the CFO was a higher transparency in the company's cost structure. Thus, the SG&A process is focused on in the following.

Max Müller described the process as follows:

> We have some interrelations from the SG&A process to other processes. However, there are not so many. The team in the Czech Republic receives a file from marketing controlling. This file serves as an input file for some of the actuals. We need the marketing costs regarding events as well as internal and external resources. However, there is the challenge that the layout of the marketing file does not fit the layout of the SG&A file. For this reason, we need to prepare the data from marketing in a different file before we can transfer them. These steps of data preparation and insertion are done by a team in the Czech Republic. Once a year, I insert the planned budget to the SG&A file. Therefore, we have somehow an interrelation to the budget file. However, the SG&A file is an input for other files such as the FC global template file as well. This Excel file is the basis to determine and upload the forecast for all of our performance indicators. (Max Müller, accountant GER)

John Smith summarized the purpose of the SG&A file and process as,

> The SG&A file is about the sales and general expenses within TheCompany. It should provide transparency about the actuals, the forecast, and the planned budget. Nevertheless, the question is: Is this level of detail sufficient? Are the breakdowns adequate? (John Smith, accountant US)

The SG&A file included general and administrative as well as sales expenses and country-specific service level agreements (SLA). However, the consultant identified that each accountant followed their own logic, resulting in a huge file—a list of more than 160 rows of costs split by columns into monthly, quarterly, and yearly values. For each month, the accountants determined a forecast. However, the monthly forecast was overwritten by the respective actuals, as soon as they were available. Therefore, there was no transparency regarding the historical development of values. Even though the German accountant was responsible for the process, there were process variants he was not aware of in detail:

> I am in charge of the SG&A file, however, the employees from the different countries are responsible for their data. You could say it is a shared responsibility where I am the global head. I am not quite sure how the others add their actuals and their forecasts. Please ask them how they do it exactly. They might pursue another approach. (Max Müller, accountant GER)

Besides the accountants, other employees and the management had a vested interest in the process:

> There are different stakeholders involved, for example, our CFO, CEO, and the head of marketing. However, most of the stakeholders have a direct link to the finance team –as I said, the CFO or the finance employees within the different countries. The team in the Czech Republic is another party involved. It prepares and inserts the actuals for Germany. For Germany, I determine the monthly forecast based on the

> actuals. John, Anna, and Karl do the same, but for their specific countries. (Max Müller, accountant GER)

To understand the global process variants and to develop an as-is model of the SG&A process, the consultant interviewed everyone involved in the process. The German accountant explained the monthly process from his perspective:

> The process starts with a new period. Then I prepare the new file, which includes the data copy from the previous months. For Germany, the team in the Czech Republic exports the data from SAP and inserts the data into the excel file. The team prepares the SAP data within different files. There, the colleagues insert the data from SAP and allocate the SG&A categories to the different items based on a code number. The code numbers are linked to cost centers. Often, there are code numbers that cannot be allocated to the categories we previously had defined. These undefined costs are put into category 66. This category code represents the costs of the financial team. We have a blur within this category. When the team of the Czech Republic is ready, I add the budget and the exchange rates, and the other countries add their actuals. When all employees have inserted their data, the team of the Czech Republic checks the overall actuals against RepSys, and I determine the forecast for Germany. I look at the past and current developments. Based on the developments and my expertise I determine the forecast for the different categories. I want to work as transparently as possible. Therefore, I do all my calculations within this file. The other countries do it on their own, whereas, everyone has their own approach. These steps are done monthly. Quarterly, I determine the SG&A allocation for all countries. (Max Müller, accountant GER)

From the US perspective, the process worked a bit different:

> In general, Max is responsible for the SG&A file. For instance, he adds the exchange rates. However, the file sometimes is not prepared when I need it. Then, I just copy it by myself and share it via SharePoint. When actuals within SAP are final and closed, I add them as a basis for the forecast. It is mostly on U+2. I extract the data from SAP and insert it into a preparation file. The prep. file has the same layout as the SG&A file. It is automatized, which means the categories are linked. Therefore, I can easily copy the data afterward. At U+5, we have to upload our forecast in RepSys. Therefore, the FC needs to be within the SG&A file then. I see it as our CFO: we need to update our actuals first and, afterward, determine the forecast. For this reason, I wait with the update until the actuals are closed and cannot change anymore. For the forecast, I have another preparation file. I think every country determines the forecast on its own. This belongs to the expertise of us as controllers. I think in the other countries my colleagues do it in a different way. (John Smith, accountant US)

In AT, the process was performed manually:

> I have to do everything manually. Absolutely manually. I cannot export anything. This is caused by SAP. Therefore, I open SAP and my own SG&A file. Then, I search for the value in SAP that I need, remember the value, and add it to the respective category within my file. Afterward, I look for the next one. I do the forecast every quarter only. Nevertheless, in the SG&A file, I have to add it every month. Therefore, I divide my forecast by three. Adding the forecast every month is micromanagement for me. I don't need that. (Karl Wagner, accountant AT)

The accountant from the NL welcomed the idea to set up a new process and report:

> I think it's very good that we rework the process and the file. Currently, I do not see any advantages in adding my data. I have only so few costs in my country. They do not make any difference in the overall picture. But a rework is very good. (Anna de Vries, accountant NL)

While the actuals came from SAP, the FC determination was based on the expertise of the accountants. Extracted from the data of previous months, the FC for each of the categories was estimated every month. However, the NL did not hand in their FC. None of the employees could assess their time efforts for adding the monthly actuals or for the forecast determination.

Based on the knowledge gained from the interviews, secondary sources, and conversations with the stakeholders, the consultant was able to create a management-oriented summary of the current process variants—therein the WS, an as-is process model of the SG&A process visualized by BPMN, and a process evaluation. The process assessment was based on the following questions—Are the stages within the process, the responsibilities, and control instances well defined? Are the levels of process functionality, the usability of the SG&A file and the data quality appropriate?

As a next step, the CFO planned a face-to-face meeting of the global team in the form of a workshop at the end of October 2018 to discuss current topics. One main part of the agenda was the business–IT alignment within the finance department. Within the slot, the consultant presented the current results and provided a proposal of a common category split of expenses, which had been worked out with Max Müller and Lena Weber before. In addition, the accountant from the US was asked to present his process due to his high degree of transparency and his first approaches to automation. The presentation of the results, the category proposal, and the process of the US served as a basis and opened a discussion for process improvements and requirement definition. *The consultant asked the accountants for their expertise and ideas for a rework to increase their involvement.* The goal was to synthesize the international process variants into one archetype process, serving as a new standard in the future:

> I have to say that some of the process steps do not work that well. We have got big differences in the approaches to how the

> data is transferred to the SG&A file and how the forecast is determined. Especially, the manual data transfer is prone to errors. Due to the different excel files and the manual data transfer, it is almost impossible to trace where possible errors have occurred. With the current SG&A file we have overshot the mark. In my opinion, we should rework the entire SG&A process and file. It has grown to a monstrous file, with a very high level of detail, various tabs, different calculations, and to some extent with hard-coded data. The structure is not clear anymore and I have no guarantee, that the numbers I copy are the correct ones, because everyone can make changes within the files. So, our system is very prone to errors: Across the countries, we neither have a common approach nor common definition of the SG&A categories, thereby, there is a risk of different interpretations and analyses depending on the employee's comprehension. (Max Müller, accountant GER)

> For me, the process works quite well. However, if we want to have the process clean, we have to redefine the categories and levels of detail needed. For me, the level of detail is the big driver of everything. If we need a more detailed view, an enterprise resource planning is necessary for the service level agreements. However, the question is, is this level of detail necessary? For me the current level is detailed enough. (John Smith, accountant US)

> First, we should commit to a common timeline and the information needed. However, let us talk about the different categories, too. The current ones are confusing and have grown. You cannot compare anything. Everyone just put his or her data in and that's it. (Anna de Vries, accountant NL)

> As I am new in the finance department, I see the process more as an external. It has grown, and we do not have any possibility to comprehend where the data came from. It is just a hard copy data. Additionally, the forecasts are overwritten by actuals. I never know: Do I look at actuals or the forecast numbers? (Lena Weber, accountant GER)

Referring to the IT, there was a common sense that SAP fulfills the requirements, yet the focus of the consultant laid on making the full use of SAP's features and on the process automation. Moreover, the consultant pointed out that the data are stored completely in SAP; hence, there was no need to rely on the marketing team and their file. Thus, the marketing file should not be part of the input data anymore. *Even if there were different opinions during the discussion, an agreement on the main points could be made*:

1. Agreement on common timeline, responsibilities, and introduction of a password governance to reduce errors
2. A common understanding of a new process: first, the process starts with a new period; second, the accountant exports of the country-specific raw data from SAP to the country-specific raw data tab within the SG&A file; third, an automatized mapping links the items (actuals) to the defined categories, based on the individual SAP code number; fourth, country-specific data check; fifth, country-specific FC determination and automatized FC calculation per month; sixth, FC freeze by Lena Weber. Apart from that, the quarterly cost allocation remains the same
3. Agreement on a new category split:
    - General expenses, including costs regarding (a) the departments (management, human resources, and finance), such as payroll and travel, (b) other employee-related costs, such as training, onboarding, and office supplies, and (c) other expenses, including audit costs or non-employee-related costs
    - Sales expenses, including expenses referring the departments (a) marketing, (b) sales, and (c) business development split into payroll, travel, events, and non-employee-related costs

The implementation started after the workshop, at the beginning of November in 2018. The CFO assessed the situation after the workshop as follows:

> Regarding the SG&A and forecasting process I do not think there will be resistance behavior. The analysis was presented at our workshop and we had a high agreement. Regarding more complex reporting and depth structure, of course, there could be resistance due to higher demand, workload and understanding. (William Jones, CFO)

During the implementation, the consultant arranged individual online meetings through Skype for Business. Therein, country-specific solutions for the mapping were worked out and the process understanding was sharpened. Due to the different SAP systems, the mapping logic developed by the consultant was individual for each country and partly very complex.

Even though the CFO did not expect user resistance from the employees, the change was not easy for everyone. While interacting with the employees individually, the consultant was confronted with different reactions. The employee from the NL was waiting and observing. The consultant gave the person the time and first worked together with employees actively supporting the change. Other employees actively shared their doubts concerning the change:

> I do not want to have a file I do not understand anymore. We had a lot of work with defining categories and building up this file. When you change something in a way we disagree, we will use a workaround. (supporting team, Czech Republic)

Another employee tried to collaborate; however, challenges due to missing skills appeared. In addition, the employee emphasized that the manual process was easier before. However, Lena Weber and the consultant worked intensively with the employee together to find a suitable

solution. With this, Karl Wagner, who had previously expressed doubts, said,

> "I'm afraid that it takes me more time to analyze the logic to understand what's behind it than entering the numbers manually every month" reconsidered and thanked for their efforts: "Thank you for your efforts! It looks very good and works easily." (Karl Wagner, accountant AT)

The consultant attached great importance on a close collaboration with the employees to get rid of the last doubts, create transparency, and establish the process as a new standard. These efforts were mirrored in more than 200 e-mails and at least one individual meeting per day. During the initiative, the CFO was kept into the loop by ad hoc and monthly update meetings. Compared to the previous SG&A process, the new one provided greater transparency of TheCompany's cost structure. On the one hand, the data were provided on a global view, while, on the other hand, every country had its own report. Based on the common categories, comparisons between the countries and over time were possible. Moreover, a chronological sequence of actuals and forecasts was comprehensible, and the process was less error-prone. In February 2019, the SG&A process was handed over from the consultant to the German accountant, who still was the global person in charge. At the end of March, the consultant left TheCompany. Until then, an advisory function was fulfilled regarding the SG&A process and another process was set into focus.

Summary

At TheCompany, after the collaboration ended, the CFO and the consultant had a concluding conversation and talked about the progress made and open topics. In total, the CFO saw many opportunities caused by the divisional change and a promising future for TheCompany. The consultant encouraged the CFO to continue with the process management as suggested in the BPM cycle. In particular, the CFO summarized,

> We have been using the new process for several weeks now. It is all going well, and I am excited to see how we managed to do this in a well-coordinated way, even without an IT department. We were able to increase automation and standardization of the SG&A process, as all employees now use the same process. We have also increased the comparability of country costs and thus reduced the number and frequency of errors in cost reporting, e.g. by increasing IT know-how of our employees. (William Jones, CFO)

The CFO, thereby, complemented about next steps within the company:

> While we had a great start in the project, we want to further advance our alignment initiative and we discuss changing business units within TheCompany. (William Jones, CFO)

Tasks for students

Please familiarize yourself with the theoretical concepts (see theoretical background). For further readings, the recommended concepts are listed in the reading list (Table 1). Afterward, read the teaching case briefly.

The case can be discussed in five rounds of analysis. Each round is designed for a processing time of 20–45 min. The initial round starts with the tasks related to the business analysis and strategy choice of TheCompany at the firm-level (30 min):

- **Task 1:** How do the given theoretical concepts interrelate to each other? Describe their relations.
- **Task 2:** Use the SAM according to Henderson and Venkatraman (1993) to analyze the chosen cross-domain alignment paths of TheCompany. Which of the four paths did TheCompany follow? Justify your answer using the four domains of the SAM.

The second part is about the business process and change management (20 min):

- **Task 3:** Did the consultant follow the BPM life cycle provided by Dumas et al. (2013)? Discuss which steps were dealt with in more in detail and which in less.
- **Task 4:** During change processes, user resistance among employees arose. Describe the different behaviors and classify the employees following Bovey and Hede (2001).

In the third part of the analysis, the focus lies on the two practical approaches—the WS and process description and visualization (45 min):

- **Task 5:** Use WST provided by Alter to create two snapshots: the first snapshot should capture the as-is of the SG&A WS and the second one, the to-be WS.
- **Task 6:** Based on WST and the process descriptions of the employees, visualize the SG&A process in one as-is and one to-be process model using BPMN. Discuss how the WS and the process have changed.

The fourth part is about process improvements (30 min):

- **Task 7:** Evaluate the as-is SG&A process based on the following criteria. Are the stages within the process, the responsibilities, and control instances well defined? How do you evaluate the process functionality, the usability of the SG&A file, and the data quality? Describe the embeddedness of the SG&A process in other processes.
- **Task 8:** Referring to your evaluation, work out a plan of fits and misfits. Where do you see room for

improvement? What would you change if you were the consultant?

Finally, close the teaching case by taking a critical look at the provided concepts (20 min):

- **Task 9:** What further short- and long-term suggestions could the consultant offer to the management TheCompany? What are specific challenges in the case for small- and medium-sized organizations? How would the recommendations change in a firm with a dedicated IT unit?
- **Task 10:** Where do you see room for improvements within the theoretical concepts and areas of further research, respectively?

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Note

1. For privacy reasons, the names of the CFO and the employees have been changed.

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